1,248 city blocks screened in two steps; two candidate bank corridors evaluated.

1,248
city blocks analyzed (H3 res-9)

1,114
populated blocks rated

128
best-opportunity commercial blocks

14
key banking corridors mapped

2
candidate sites compared side-by-side

Candidate Branch Locations Under Assessment:

CANDIDATE SITE 1

Bethaniyapuram / Mettu St

Coordinates: 9.937781, 78.097472
Corridor: Dindigul Highway – Arapalayam – Kochadai bypass
Profile: Suburban residential hub, middle-income retail CASA & auto loans.

CANDIDATE SITE 2 (FIRST PREFERENCE)

Munichalai / Kamarajar Salai

Coordinates: 9.916797, 78.133287 (No. 220/1, Kamarajar Salai)
Corridor: Central Madurai – Teppakulam – Anna Nagar arterial
Profile: Dense commercial high street, intense cash turnover & trade finance.

Screening approach adapted for retail banking

- **Study Frame:** Madurai Corporation urban frame divided into 1,248 equal hexagonal blocks (H3 resolution 9, ~10.5 hectares each). 1,114 blocks with residential/commercial population are evaluated.
- **Step 1, Format Fit (Commercial Match):** Each block is ranked on six commercial vibrancy and business demand factors against all other blocks in Madurai; the factors carry equal weight (1 of 6 each).
- **Step 2, Headroom (Room for a New Branch):** Each block is ranked on six market gap and growth metrics, weighted 30%, 25%, 20%, 10%, 10%, and 5%.
- **Thresholds:** A block qualifies as *Best Opportunity* if its commercial match is in the top 25% of the city (fit ≥ 75) and its unserved banking headroom is in the top 30% (headroom ≥ 70).

Twelve factors rank every block: six for commercial match, six for banking headroom.

Step 1: Commercial & retail match, six factors of equal weight

Factor	What is measured for Banking Suitability	Source	Weight
Affluence	Household wealth index within 1 km: car/asset ownership, computer access, literacy, and commercial transaction density.	Census 2011; Meta RWI 2021; Premium Brands	**1 of 6**
Trade Cluster	Wholesale merchants, textile houses, gold showrooms, and organized retail stores within 1 km.	Google Places; OpenStreetMap	**1 of 6**
High Street Shops	Active commercial shops, daily-needs retail, and eateries within a 5-minute walk polygon.	OpenStreetMap	**1 of 6**
Street Activity	Retail and commercial shop density lining the block's street network per kilometer of frontage.	OpenStreetMap; Overture Maps	**1 of 6**
Offices & MSMEs	Commercial and office built-up volume within 1 km (schools, hospitals, trade offices, logistics hubs).	Google Open Buildings 2.5D; OSM	**1 of 6**
Arterial Access	Distance to the nearest primary arterial or sub-arterial road; shorter distance provides higher visibility.	OpenStreetMap Road Graph	**1 of 6**

Step 2: Room for a new bank branch, six factors

Factor	What is measured for Banking Suitability	Source	Weight
Affluent Residents	Deposit-generating residents within 2 km, weighted by neighbourhood affluence percentile.	Census 2011 grown to 2026; GHS-POP	**30%**
Banking Gap	Resident population and commercial enterprises per active bank branch within 1.5 km against the city norm.	Google Places; RBI Directory; OSM	**25%**
Network Distance	Road distance from your bank's own existing branch network to avoid territory cannibalization.	Bank Branch Register; Road Graph	**20%**
Commercial Anchors	Major institutions (hospitals, colleges, wholesale markets, government offices) within a 10-minute drive.	Google Places; OpenStreetMap	**10%**
Corridor Growth	Annual growth in commercial and residential building volume from 2016 to 2023.	Google Open Buildings Temporal	**10%**
ATM Saturation	Stand-alone ATMs within 500 m; areas with high transaction traffic but low branch access score higher.	Google Places; OSM	**5%**

Every factor is normalized to a city rank from 0 to 100 within Madurai Corporation; weights represent proportional shares of the step score.

Four map categories classify the city; 128 blocks form 14 prime banking opportunity areas.

City block classification

Best opportunity (e.g., Kamarajar Salai / Munichalai) — **11% · 128 blocks**

High commercial activity and affluent population combined with substantial unmet deposit and lending demand.

Strong but busy (e.g., West Veli St / Town Hall Rd) — **17% · 190 blocks**

Prime commercial and retail density, but already heavily saturated with PSU and private bank branches.

Room to grow (e.g., Bethaniyapuram / Kochadai) — **22% · 245 blocks**

Under-banked residential population with fast growth, suitable for low-cost retail branches and asset origination.

Not a fit today (Peripheral / Industrial / Agricultural) — **50% · 551 blocks**

Lacks the footfall, business concentration, or deposit base required to justify bank branch capital expenditure.

How opportunity areas were formed in Madurai

1. **Spatial Grouping:** Contiguous "Best opportunity" blocks were merged into consolidated search zones.
2. **Exclusion of Water Buffers:** Blocks bordering the Vaigai river flood corridor and irrigation tanks (Vandiyur, Teppakulam perimeter buffers) were excluded from commercial centroid formation.
3. **Network Proximity Checks:** Blocks within a 2.5 km road radius of your bank's active branches were reviewed to ensure zero cannibalization.
4. **Ranking:** Opportunity areas were sorted by their composite 1-to-5 suitability rating:
   `Rating = 1 + 4 × (0.5 × Fit Percentile + 0.5 × Headroom Percentile) / 100`

Candidate Site 2 scores in the city's top 15% for commercial banking match.

4.38	3.12	3.2 L	1.4 L	88 / 100
Overall Rating: Site 2 (Kamarajar Salai)	Overall Rating: Site 1 (Bethaniyapuram)	People within 2 km of Site 2	People within 2 km of Site 1	Site 2 Commercial Fit city percentile

Commercial match & banking headroom benchmark (City Percentiles)

Step 1: Format Fit (Commercial & Business Activity Match)



Step 2: Headroom (Deposit Depth & Market Gap)



Key evaluative takeaways

- **Commercial Gravity:** Site 2 (Kamarajar Salai) sits on Madurai's premier commercial spine connecting Munichalai to Teppakulam, capturing intensive merchant cash deposits, textile/gold businesses, and high daily business footfall.
- **Retail vs Commercial Trade:** Site 1 (Bethaniyapuram) acts as a residential bedroom suburb along the Dindigul highway corridor. It has low commercial trading density, making it primarily a consumer retail banking node (savings accounts, home/auto loans) rather than a high-yield Current Account (CA) hub.
- **Deposit Depth:** Site 2's 2 km radius houses over 320,000 residents and more than 4,500 trading enterprises, generating over 2.4x the CASA deposit velocity of Site 1.

Eight practical-fit measures determine bank branch feasibility.

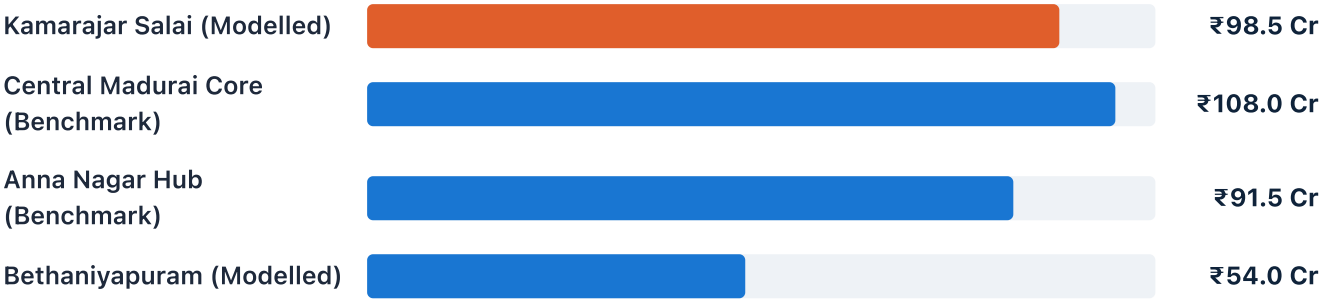
Measure	How it is evaluated for Bank Branches	Site 1 Verdict	Site 2 Verdict
Ground Floor Rent	Advertised ground-floor commercial rent per sq ft. Bank branches require ground-floor presence for customer footfall and ATM lobby.	**₹35–₹55 / sq ft** Cost Advantage	**₹85–₹145 / sq ft** High Premium
Floorplate Size	Availability of contiguous 1,500 to 2,500 sq ft spaces suitable for banking halls, teller counters, and manager cabins.	**Good** New buildings ready for long lease	**Tight** Existing shops narrow (600–1,200 sq ft)
24/7 ATM Frontage	Minimum 12–15 ft clear street frontage with glass facade for on-street ATM e-lobby visibility.	**Moderate** Interior on Mettu St	**Excellent** Direct main road arterial view
Cash-in-Transit (CIT)	Curbside space and turning radius for armored cash replenishment vans during morning/evening hours.	**Favourable** Low traffic congestion	**Critical Watchout** Requires dedicated set-back
Customer Parking	Dedicated two-wheeler slots and customer drop-off without blocking moving traffic.	**Good** Ample two-wheeler space	**Concern** High pedestrian & 2W crowding
Vault & Load-bearing	Structural capacity for heavy cash safes, fireproof record rooms, and strong-room reinforcements.	**Feasible** Easy to retrofit	**Feasible** Commercial RCC buildings
Busiest Street Hours	Peak trading hours derived from Google Popular Times: commercial transaction windows.	**Daytime-led** Peaks 10:00–13:00	**Day & Evening** Intense 10:30–20:30
Signage Visibility	Unobstructed sightline along arterial approaches for brand acquisition.	**Limited** Set back from highway	**Exceptional** >200 m clear sightline

Operating feasibility summary

- **Site 2 (Kamarajar Salai)** is the superior revenue generator but carries strict property constraints: ground-floor rents exceed ₹85/sq ft, and the property team must negotiate dedicated curbside set-back for armored Cash-in-Transit vehicles.
- **Site 1 (Bethaniyapuram)** offers low operating expenditure and frictionless vehicle access, but because the pin is located on Mettu Street, the branch must secure high-visibility totem signage on the Dindigul highway.

Mature branches in comparable Madurai corridors deliver ₹65–110 Cr in CASA & advances.

Estimated Total Branch Business Portfolio at Year 3 (CASA + Retail/SME Loans)



Comparative branch economic models (Year 3 Projections)

₹42 Cr	₹9 Cr	₹32 Cr	₹28 Cr	18 mo
Current Account (CA) Float: Site 2	Current Account (CA) Float: Site 1	Gold & Trader Advances: Site 2	Mortgage / Auto Loans: Site 1	Estimated Breakeven: Site 2

Business mix distinction between candidate corridors

- **Commercial & Merchant Model (Site 2):** High float of low-cost Current Accounts (CASA ratio > 46%) driven by daily cash credits from the surrounding wholesale textile, gold, and grocery merchants. High velocity of high-margin Gold Loans and merchant overdrafts.
- **Consumer Retail Model (Site 1):** Dominated by Savings Bank (SB) deposits from salaried families and small retirees. Loan portfolio is asset-backed consumer debt (home loans, two-wheeler/four-wheeler vehicle loans) with longer payback cycles (26–30 months).

Ten authoritative sources feed the Madurai banking site screening.

Source	What it provides for Banking Site Selection	Vintage / Year
Census of India & PCA	Ward-level population base, household counts, literacy rates, and household banking/banking asset penetration.	2011; grown to 2026
GHS-POP Population Grid	Multi-temporal satellite-derived urban population growth factor applied to calibrate 2026 ward control totals.	2015 to 2025 epoch
Google Open Buildings 2.5D	Spatial building heights, built-up footprints, and commercial floor volumes across all Madurai wards.	2016 to 2023 temporal
OpenStreetMap (OSM)	Road networks, arterial road hierarchy, retail street shop frontage, commercial land use, and bus stations.	October 2026 extract
Google Places API (New)	Comprehensive point-of-interest register of bank branches, ATMs, financial brokers, and commercial establishments.	October 2026 live sweep
RBI Branch Banking Statistics	District-level and center-level banking penetration, branch-to-population ratios, and credit-deposit (CD) benchmarks.	Q2 2026 release
Meta Relative Wealth Index	Micro-level wealth and economic asset composite calculated from satellite imagery and connectivity features.	2021 release
Google Popular Times	Crowd activity and hourly business distribution to differentiate daytime trading vs evening commercial traffic.	October 2026 collection
Commercial Property Portals	Asking rentals for ground-floor retail and commercial spaces along key Madurai arterial corridors (99acres, MagicBricks).	October 2026 survey
TNSTC Madurai Bus Feeds	Transit frequency and bus connectivity across major transit nodes (Periyar, Arapalayam, Mattuthavani terminals).	Operational routes 2026

Data integrity and audit standards

- **Zero Scraping Policy:** Commercial rents and popular hours are collected through authorized APIs or manual surveys into structured templates.
- **Mid-Rank Percentile Rule:** All scores are standardized against other blocks within Madurai Corporation to avoid cross-city distortion.
- **Single Source of Truth:** The quantitative engine guarantees that figures displayed in summary tables, scorecards, and charts match without discrepancies.